

Span composition using fake pullbacks

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Abstract

The construction of a category of spans can be made in some categories $\mathcal{C}$ which do not have pullbacks in the traditional sense. The PROP for monoids is a good example of such a $\mathcal{C}$. The 2012 book concerning homological algebra by Marco Grandis gives the proof of associativity of relations in a Puppe-exact category based on a 1967 paper of M.Š. Calenko. The proof here is a restructuring of that proof in the spirit of the first sentence of this Abstract. We observe that these relations are spans of EM-spans and that EM-spans admit fake pullbacks so that spans of EM-spans compose. Our setting is more general than Puppe-exact categories.

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Introduction

The construction of a category of spans can be made in some categories $\mathcal{C}$ not having pullbacks in the traditional sense, only having some form of *fake pullback*. The PROP for monoids is a good example of such a $\mathcal{C}$; it has a forgetful functor to the category of finite sets which takes fake pullbacks to genuine pullbacks.

As discussed in the book [8] by Marco Grandis, relations in a Puppe-exact category $\mathcal{C}$ are zig-zag diagrams of monomorphisms and epimorphisms, not just jointly monomorphic spans as for a regular category (see [5] for example). Associativity of these zig-zag relations was proved by M.Š. Calenko [10] over 50 years ago; also see [4] Appendix A.5, pages 140–142.

The present paper is a restructuring of the associativity proof in the spirit of fake pullbacks. The original category $\mathcal{C}$ does not even need to be pointed, but it should have a *suitable factorization system* $(\mathcal{E}, \mathcal{M})$. The fake pullbacks are constructed in what we call $\text{Spn}(\mathcal{E}, \mathcal{M})$, not in $\mathcal{C}$ itself, and there is no forgetful functor turning them into genuine pullbacks. The relations are spans in $\text{Spn}(\mathcal{E}, \mathcal{M})$. The main point in proving associativity of the span composition is that fake pullbacks stack properly.

1 Suitable factorization systems

Let $(\mathcal{E}, \mathcal{M})$ be a factorization system in the sense of [6] on a category $\mathcal{C}$. That is, $\mathcal{E}$ and $\mathcal{M}$ are sets of morphisms of $\mathcal{C}$ which contain the isomorphisms, are closed under composition, and satisfy the conditions:

FS1. if $mu = ve$ with $e \in \mathcal{E}$ and $m \in \mathcal{M}$ then there exists a unique w with $we = u$ and $mw = v$;

FS2. every morphism f factorizes $f = m \circ e$ with $e \in \mathcal{E}$ and $m \in \mathcal{M}$.

If we write $f : A \rightarrow B$, we mean $f \in \mathcal{E}$. If we write $f : A \rightarrowtail B$, we mean $f \in \mathcal{M}$. Another way to express FS1 is to ask, for all $X \xrightarrow{e} Y \in \mathcal{E}$ and $A \xrightarrow{m} B \in \mathcal{M}$, that the square (1.1) should be a pullback.

$$\begin{array}{ccc} \mathcal{C}(Y, A) & \xrightarrow{\mathcal{C}(e, A)} & \mathcal{C}(X, A) \\ \mathcal{C}(Y, m) \downarrow & & \downarrow \mathcal{C}(X, m) \\ \mathcal{C}(Y, B) & \xrightarrow{\mathcal{C}(e, A)} & \mathcal{C}(X, B) \end{array} \quad (1.1)$$

Remark 1. If we were dealing with a factorization system on a bicategory $\mathcal{C}$, we would ask (1.1) (with the associativity constraint providing a natural isomorphism in the square) to be a bipullback. Also, in FS2, we would only ask $f \cong m \circ e$. This is relevant to Proposition 6 and Section 5 below.

The factorization system $(\mathcal{E}, \mathcal{M})$ is *suitable* when it satisfies:

- SFS1. pullbacks of arbitrary morphisms along members of $\mathcal{M}$ exist;
- SFS2. pushouts of arbitrary morphisms along members of $\mathcal{E}$ exist;
- SFS3. the pullback of an $\mathcal{E}$ along an $\mathcal{M}$ is in $\mathcal{E}$;
- SFS4. the pushout of an $\mathcal{M}$ along an $\mathcal{E}$ is in $\mathcal{M}$;
- SFS5. a commutative square of the form

$$\begin{array}{ccc} D & \xrightarrow{n} & B \\ e \downarrow & & \downarrow d \\ A & \xrightarrow{m} & C \end{array}$$

is a pullback if and only if it is a pushout.

Proposition 2. (i) *Spans of the form $X \leftarrow S \rightarrow Y$ are jointly monomorphic.*
(ii) *Cospans of the form $X \rightarrow C \leftarrow Y$ are jointly epimorphic.*

Proof. A pullback of $X \rightarrow C \leftarrow Y$ exists by SFS1 and $X \rightarrow C \leftarrow Y$ is the pushout of the resultant span by SFS5. Pushout cospans are jointly epimorphic. This proves (ii), and (i) is dual. $\square$

- Example 1.**
1. Take $\mathcal{C}$ to be the category Grp of groups, $\mathcal{E}$ to be the set of surjective morphisms and $\mathcal{M}$ to be the set of injective morphisms.
 2. Take $\mathcal{C}$ to be any Puppe-exact category as studied by Grandis [8], $\mathcal{E}$ the epimorphisms and $\mathcal{M}$ the monomorphisms. This includes all abelian categories.
 3. Take $\mathcal{C}$ to be the category $\text{Spn}[\text{Set}_{\text{inj}}]$ of spans in the category of sets and injective functions, $\mathcal{E}$ the i^* and $\mathcal{M}$ the i_* .
 4. Take $\mathcal{C}$ to be any groupoid with $\mathcal{E} = \mathcal{M}$ containing all morphisms.

Now we remind the reader of Lemma 2.5.9 from [8].

Lemma 3. *In a commutative diagram of the form*

$$\begin{array}{ccccc} A & \xrightarrow{d} & B & \xrightarrow{i} & C \\ \ell \downarrow & & \downarrow m & & \downarrow n \\ X & \xrightarrow{e} & Y & \xrightarrow{j} & Z \end{array}, \quad (1.2)$$

the horizontally pasted square is a pullback if and only if both the component squares are pullbacks.

Proof. “If” is true without any condition on the morphisms. For the converse, using SFS1, take the pullback of j and n to obtain another pastable pair of squares with the same left, right and bottom sides. The top composites are equal. By factorization system properties and SFS3, the new top is also a factorization of $i \circ d$ and thus isomorphic to the given factorization. So both of the old squares are also pullbacks. $\square$

We might call the diagram (1.2) an $\mathcal{M}$ -morphism of factorizations. The dual of the lemma concerns pushouts in $\mathcal{E}$ -morphisms of factorizations; it also holds since we did not use SFS5 in proving the Lemma. However condition SFS5 does tell us that the left square of (1.2) is also a pushout when the pasted diagram is a pullback.

2 The bicategory of EM-spans

Some terminology used here, for bicategories, spans and discrete fibrations, is explained in [9].

Let $(\mathcal{E}, \mathcal{M})$ be a suitable factorization system on the category $\mathcal{C}$.

We define a bicategory $\text{Spn}(\mathcal{E}, \mathcal{M})$ with the same objects as $\mathcal{C}$. The morphisms $(d, R, m) : U \rightarrow W$ are spans $U \leftarrow R \rightarrow W$ in $\mathcal{C}$. The 2-cells are the usual morphisms of spans. Composition is the usual composition of spans; this uses conditions SFS1, SFS3 and closure of $\mathcal{E}$ under composition.

Each $(X \xrightarrow{m} Y) \in \mathcal{M}$ gives a morphism $m_* : X \xrightarrow{(1_X, X, m)} Y$ in $\text{Spn}(\mathcal{E}, \mathcal{M})$ and each $(X \xrightarrow{e} Y) \in \mathcal{E}$ gives a morphism $e^* : Y \xrightarrow{(e, X, 1_X)} X$ in $\text{Spn}(\mathcal{E}, \mathcal{M})$. Write $\mathcal{M}_*$ for the class of all morphisms isomorphic to m_* for some $m \in \mathcal{M}$ and write $\mathcal{E}^*$ for the class of all morphisms isomorphic to e^* with $e \in \mathcal{E}$.

Notice that 2-cells between members of $\mathcal{M}_*$, 2-cells between members of $\mathcal{E}^*$, and 2-cells from a member of $\mathcal{E}^*$ to a member of $\mathcal{M}_*$, are all invertible.

Proposition 2 tells us that the bicategory $\text{Spn}(\mathcal{E}, \mathcal{M})$ is locally preordered.

Proposition 4. *Given $X \xrightarrow{m_*} Y \in \mathcal{M}_*$ and $Z \xrightarrow{e^*} Y \in \mathcal{E}^*$, there exists a diagram of the form*

$$\begin{array}{ccc} J & \xrightarrow{\bar{e}^*} & X \\ \bar{m}_* \downarrow & \Leftarrow & \downarrow m_* \\ Z & \xrightarrow{e^*} & Y \end{array} \quad (2.3)$$

in $\text{Spn}(\mathcal{E}, \mathcal{M})$, with $\bar{e} \in \mathcal{E}$ and $\bar{m} \in \mathcal{M}$, which is unique up to isomorphism.

Proof. Interpreting $m_* \circ \bar{e}^* \leq e^* \circ \bar{m}_*$, we see that $\bar{m} \circ \bar{e}$ is forced to be an $(\mathcal{E}, \mathcal{M})$ factorization of $e \circ m$. $\square$

Proposition 5. *If $m \in \mathcal{M}$ then m_* is a discrete fibration in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$; that is, each functor $\mathrm{Spn}(\mathcal{E}, \mathcal{M})(K, m_*)$ is a discrete fibration.*

Proposition 6. *$(\mathcal{E}^*, \mathcal{M}_*)$ is a factorization system on the bicategory $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$.*

Proof. Every morphism $(d, R, m) : U \rightarrow W$ decomposes as $U \xrightarrow{d^*} R \xrightarrow{m_*} W$; this decomposition $\mathcal{M}_* \mathcal{E}^*$ is unique up to isomorphism. The bipullback form of FS1 can be readily checked for this factorization. $\square$

Proposition 7. *Pullbacks in $\mathcal{C}$ whose morphisms are all in $\mathcal{M}$ are taken by $(-)_*$ to bipullbacks in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$. Also, pushouts in $\mathcal{C}$ whose morphisms are all in $\mathcal{E}$ are taken by $(-)^*$ to bipullbacks in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$.*

3 Relations as spans of spans

By regular categories we mean those in the sense of Barr [1] which admit all finite limits. One characterization of the bicategory of relations in a regular category was given in [5]. A relation from X to Z in a regular category is a jointly monomorphic span from X to Z ; these are composed using span composition followed by factorization. Equivalently, a relation from X to Z is a subobject of $X \times Z$.

The category Grp of groups is regular. So relations are subgroups of products $X \times Z$. The Goursat Lemma [7] is a bijection between subgroups $S \leq X \times Z$ of a cartesian product of groups X and Z and end-fixed isomorphism classes of diagrams

$$X \xleftarrow{m} U \xrightarrow{d} Y \xleftarrow{e} V \xrightarrow{n} Z . \quad (3.4)$$

To obtain S from (3.4), take the pullback $U \xleftarrow{\bar{e}} P \xrightarrow{\bar{d}} V$ of $U \xrightarrow{d} Y \xleftarrow{e} V$ then S is the image of $P \xrightarrow{(m\bar{e}, e\bar{d})} X \times Y$. To obtain the zig-zag (3.4) from $S \hookrightarrow X \times Z$, factorize the two restricted projections to obtain

$$X \xleftarrow{m} U \xleftarrow{e'} S \xrightarrow{d'} V \xrightarrow{n} Z ,$$

then pushout e' and d' to obtain d and e .

This motivates the definition of *relation* from X to Z in a category $\mathcal{C}$ equipped with a suitable factorization system $(\mathcal{E}, \mathcal{M})$ as an isomorphism class of diagrams of the form (3.4). A good reference is [8] for the case where $\mathcal{C}$ is Puppe-exact.

The starting point for the present paper was the simple observation that a relation diagram (3.4) is a span in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$:

$$(d, U, m, Y, e, V, n) : X \xleftarrow{(d, U, m)} Y \xrightarrow{(e, V, n)} Z . \quad (3.5)$$

Write $\mathrm{Spn}[\mathcal{E}, \mathcal{M}]$ for the classifying category of the bicategory $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$; it has the same objects as $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$ and isomorphism classes $[e, S, m]$ of morphisms (e, S, m) . We would like to define the bicategory $\mathrm{Rel}(\mathcal{E}, \mathcal{M})$ to be $\mathrm{Spn}(\mathrm{Spn}[\mathcal{E}, \mathcal{M}])$. This is satisfactory as a definition of the 2-graph and vertical composition, but for the horizontal composition we need a way to compose spans in $\mathrm{Spn}[\mathcal{E}, \mathcal{M}]$.

4 A fake pullback construction

Let $(\mathcal{E}, \mathcal{M})$ be a suitable factorization system on a category $\mathcal{C}$. Although $\mathcal{C}$ may not have all pullbacks, we will now show that $\text{Spn}[\mathcal{E}, \mathcal{M}]$ does allow some kind of span composition and this gives a composition of relations. The construction and proof of associativity restructures that of [10]. We will see in Section 5 that the properties of $\text{Spn}(\mathcal{E}, \mathcal{M})$ established in Section 2 allow an abstract proof of associativity of composition of relations.

Take any cospan $U \xrightarrow{(d,R,m)} W \xleftarrow{(e,S,n)} V$ in $\text{Spn}(\mathcal{E}, \mathcal{M})$. Construct the diagram

$$\begin{array}{ccccc}
 Q & \xleftarrow{s} & Y & \xrightarrow{j} & V \\
 \uparrow r & & \uparrow \bar{e} & & \uparrow e \\
 X & \xleftarrow{\bar{d}} & Z & \xrightarrow{\bar{m}} & S \\
 \downarrow i & & \downarrow \bar{n} & & \downarrow n \\
 U & \xleftarrow{d} & R & \xrightarrow{m} & W
 \end{array} \tag{4.6}$$

in which the bottom right square is a pullback of $R \rightrightarrows W \leftarrow S$, the bottom left square is an $(\mathcal{E}, \mathcal{M})$ -factorization of the composite $Z \rightrightarrows R \rightarrow U$, the top right square is an $(\mathcal{E}, \mathcal{M})$ -factorization of the composite $Z \rightrightarrows S \rightarrow V$, and the top left square is a pushout of the span $X \leftarrow Z \rightarrow Y$.

We call the span $U \xleftarrow{(r,R,i)} Q \xrightarrow{(s,S,j)} V$ the *fake pullback* of the given cospan $U \xrightarrow{(d,R,m)} W \xleftarrow{(e,S,n)} V$. We obtain the diagram (4.7) in $\text{Spn}(\mathcal{E}, \mathcal{M})$. The top left square comes from a pushout in $\mathcal{C}$, the bottom right square from a pullback in $\mathcal{C}$, while the 2-cells come from factorizing an $\mathcal{M}$ followed by an $\mathcal{E}$ as an $\mathcal{E}$ followed by an $\mathcal{M}$.

$$\begin{array}{ccccc}
 Q & \xrightarrow{s^*} & Y & \xrightarrow{j^*} & V \\
 r^* \downarrow & \lrcorner & \downarrow \bar{e}^* & \Rightarrow & \downarrow e^* \\
 X & \xrightarrow{\bar{d}^*} & Z & \xrightarrow{\bar{m}_*} & S \\
 i_* \downarrow & \Leftarrow & \downarrow \bar{n}_* & \lrcorner & \downarrow n_* \\
 U & \xrightarrow{d^*} & R & \xrightarrow{m_*} & W
 \end{array} \tag{4.7}$$

Remark 8. a. If d is invertible, so is s . If m is invertible, so is j .

b. If $(\mathcal{E}, \mathcal{M})$ is proper (that is, every $\mathcal{E}$ is an epimorphism and every $\mathcal{M}$ is a monomorphism) then every morphism $\mathbf{r} : X \rightarrow Y$ of $\text{Spn}(\mathcal{E}, \mathcal{M})$ is a “fake monomorphism” in the sense that the fake pullback of $X \xrightarrow{\mathbf{r}} Y \xleftarrow{\mathbf{r}} X$ is the identity span $X \xleftarrow{1_X} X \xrightarrow{1_X} X$.

5 An abstraction

A bicategory $\mathcal{S}$ is defined to be *fake pullback ready* when it is locally preordered and is equipped with a factorization system $(\mathcal{U}, \mathcal{L})$ satisfying the following conditions:

- V1. bipullbacks of $\mathcal{U}$ s along $\mathcal{U}$ s exist and are in $\mathcal{U}$, and bipullbacks of $\mathcal{L}$ s along $\mathcal{L}$ s exist and are in $\mathcal{L}$;
- V2. given $X \xrightarrow{a} Z \xleftarrow{x} Y$ with $a \in \mathcal{U}$ and $x \in \mathcal{L}$, there exists a square

$$\begin{array}{ccc} U & \xrightarrow{b} & Y \\ y \downarrow & \Leftarrow & \downarrow x \\ X & \xrightarrow{a} & Z, \end{array} \quad (5.8)$$

with $b \in \mathcal{U}$ and $y \in \mathcal{L}$, which is unique up to equivalence;

- V3. given a diagram

$$\begin{array}{ccccc} X & \xrightarrow{x} & Y & \xrightarrow{a} & Z \\ r \downarrow & \lrcorner & s \downarrow & \Leftarrow & \downarrow t \\ A & \xrightarrow{y} & B & \xrightarrow{b} & C, \end{array} \quad (5.9)$$

with the left square a bipullback, $r, s, t, x, y \in \mathcal{L}$ and $a, b \in \mathcal{U}$, and factorizations $a \circ x \cong v \circ c$ and $b \circ y \cong w \circ d$ with $v, w \in \mathcal{L}$ and $c, d \in \mathcal{U}$, there exists a diagram

$$\begin{array}{ccccc} X & \xrightarrow{c} & I & \xrightarrow{v} & Z \\ r \downarrow & \Leftarrow & \downarrow q & \lrcorner & \downarrow t \\ A & \xrightarrow{d} & J & \xrightarrow{w} & C \end{array} \quad (5.10)$$

with the right square a bipullback and $q \in \mathcal{L}$;

- V4. given a diagram

$$\begin{array}{ccccc} D & \xrightarrow{u} & E & \xrightarrow{h} & F \\ e \downarrow & \Rightarrow & f \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{x} & Y & \xrightarrow{a} & Z \end{array} \quad (5.11)$$

with the right square a bipullback, $x, u \in \mathcal{L}$ and $h, a, e, f, g \in \mathcal{U}$, and factorizations $h \circ u \cong p \circ k$ and $a \circ x \cong v \circ c$ with $v, p \in \mathcal{L}$ and $c, k \in \mathcal{U}$, there exists a diagram

$$\begin{array}{ccccc} D & \xrightarrow{k} & K & \xrightarrow{p} & F \\ e \downarrow & \lrcorner & \downarrow j & \Rightarrow & \downarrow g \\ X & \xrightarrow{c} & I & \xrightarrow{v} & Z \end{array} \quad (5.12)$$

with the left square a bipullback and $j \in \mathcal{U}$.

Proposition 9. *Let $(\mathcal{E}, \mathcal{M})$ be a suitable factorization system on the category $\mathcal{C}$. The locally preordered bicategory $\text{Spn}(\mathcal{E}, \mathcal{M})$ is rendered fake pullback ready by the factorization system $(\mathcal{E}^*, \mathcal{M}_*)$ of Proposition 6.*

Proof. Condition V1 is provided by Proposition 7. Condition V2 is provided by Proposition 4. Consider diagram (5.9) with $x_*, a^*, \dots$ replacing $x, a, \dots$ since $\mathcal{L} = \mathcal{M}_*$ and $\mathcal{U} = \mathcal{E}^*$ in this case. The left square amounts to the pullback shown as the right-hand square on the left-hand side of (5.13). The right-hand square with the 2-cell amounts to the factorization $s \circ a = b \circ t$. Now form the pullback on the left of the left-hand side of (5.13) and the pullback on the right of the right-hand side of (5.13). Since $btv = sav = sxc = yrc$, there exists a unique q such that $dq = rc$ and $wq = tv$. So we have the equal pastings as shown in (5.13).

$$\begin{array}{ccc}
 I & \xrightarrow{c} & X & \xrightarrow{r} & A \\
 \downarrow v & \lrcorner & \downarrow x & \lrcorner & \downarrow y \\
 Z & \xrightarrow{a} & Y & \xrightarrow{s} & B
 \end{array}
 =
 \begin{array}{ccc}
 I & \xrightarrow{q} & J & \xrightarrow{d} & A \\
 \downarrow v & & \downarrow w & \lrcorner & \downarrow y \\
 Z & \xrightarrow{t} & C & \xrightarrow{b} & B
 \end{array}
 \quad (5.13)$$

It follows that the left diagram on the right-hand side of (5.13) is a pullback and, by SFS3, that $q \in \mathcal{E}$. Diagram (5.10) results.

It is V4 which requires suitable factorization condition SFS5. Consider diagram (5.11). We have the pushout on the right of the left-hand side of (5.14) and the factorization $fx = ue$. Form the pullback of a and x and note, using one direction of SFS5, that it gives the pushout on the left of the left-hand side of (5.14). Next, factorize $gv = pj$ through K with $p \in \mathcal{M}$ and $j \in \mathcal{E}$. Using functoriality of factorization FS1, we obtain a unique $k : K \rightarrow D$ with $kj = ec$ and $uk = hp$.

$$\begin{array}{ccc}
 I & \xrightarrow{v} & Z & \xrightarrow{g} & F \\
 \downarrow c & \lrcorner & \downarrow a & \lrcorner & \downarrow h \\
 X & \xrightarrow{x} & Y & \xrightarrow{f} & E
 \end{array}
 =
 \begin{array}{ccc}
 I & \xrightarrow{j} & K & \xrightarrow{p} & F \\
 \downarrow c & & \downarrow k & & \downarrow h \\
 X & \xrightarrow{e} & D & \xrightarrow{u} & E
 \end{array}
 \quad (5.14)$$

It follows that both squares on the right-hand side of (5.14) are pushouts. Diagram (5.12) results using the other direction of SFS5 to see that the right square on the right-hand side of (5.14) is a pullback and hence $p_*k^* = h^*u_*$. $\square$

Let $\mathcal{S}$ be fake pullback ready. The *fake pullback* of a cospan $U \xrightarrow{r} W \xleftarrow{s} V$ in $\mathcal{S}$ is constructed as follows. Factorize $r \cong x \circ a$ and $s \cong y \circ b$ with $a, b \in \mathcal{U}$ and $x, y \in \mathcal{L}$. Using half of V1, take the bipullback of x and y as shown in the bottom right square of (5.15). Now construct the bottom left and top right squares

of (5.15) using V2. Using the other half of V1, we obtain the top left bipullback.

$$\begin{array}{ccccc}
 Q & \xrightarrow{a'} & Y & \xrightarrow{\bar{x}} & V \\
 b' \downarrow & \lrcorner & \downarrow b & \Rightarrow & \downarrow b \\
 X & \xrightarrow{\bar{a}} & Z & \xrightarrow{x'} & S \\
 \bar{y} \downarrow & \Leftarrow & \downarrow y' & \lrcorner & \downarrow y \\
 U & \xrightarrow{a} & R & \xrightarrow{x} & W
 \end{array} \quad (5.15)$$

The span $U \xleftarrow{\bar{y}b'} Q \xrightarrow{\bar{x}a'} V$ is our fake pullback of $U \xrightarrow{r} W \xleftarrow{s} V$.

Proposition 10. *Fake pullbacks are symmetric. That is, if $U \xleftarrow{\bar{s}} Q \xrightarrow{\bar{r}} V$ is a fake pullback of $U \xrightarrow{r} W \xleftarrow{s} V$ then $V \xleftarrow{\bar{r}} Q \xrightarrow{\bar{s}} U$ is a fake pullback of $V \xrightarrow{s} W \xleftarrow{r} U$.*

Proof. In (5.15), the bipullbacks are symmetric and both 2-cells point to the boundary of the diagram. So the diagram is symmetric about its main diagonal. $\square$

Note that, should a bipullback

$$\begin{array}{ccc}
 U & \xrightarrow{b} & Y \\
 y \downarrow & \Leftarrow & \downarrow x \\
 X & \xrightarrow{a} & Z
 \end{array},$$

of $X \xrightarrow{a} Z \xleftarrow{x} Y$ exist with $y \in \mathcal{L}$ and $b \in \mathcal{U}$, it would provide the square for V2. This happens for example when a is an identity, b is an identity, and $y = x$. Consequently:

Proposition 11. *An identity morphism provides a fake pullback of an identity morphism along any morphism.*

$$\begin{array}{ccc}
 P & \xrightarrow{\bar{t}} & Q & \xrightarrow{\bar{r}} & V \\
 \bar{s} \downarrow & \text{fakepb} & \downarrow \bar{s} & \text{fakepb} & \downarrow s \\
 X & \xrightarrow{t} & U & \xrightarrow{r} & W
 \end{array} \quad \begin{array}{ccc}
 P & \xrightarrow{\bar{r}\bar{t}} & V \\
 \bar{s} \downarrow & \text{fakepb} & \downarrow s \\
 X & \xrightarrow{rt} & W
 \end{array} \quad (5.16)$$

Proposition 12. *Fake pullbacks stack. That is, if the two squares on the left of (5.16) are fake pullbacks then so is the pasted square on the right of (5.16).*

Proof. Faced with a diagram like

$$\begin{array}{ccccccc}
Q & \xrightarrow{u} & Y & \xrightarrow{\ell} & V & \xrightarrow{u} & A \xrightarrow{\ell} D \\
\downarrow u & \lrcorner & \downarrow u & \Rightarrow & \downarrow u & \lrcorner & \downarrow u \\
X & \xrightarrow{u} & Z & \xrightarrow{\ell} & S & \xrightarrow{u} & B \xrightarrow{\ell} E \\
\downarrow \ell & \Leftarrow & \downarrow \ell & \lrcorner & \downarrow \ell & \Leftarrow & \downarrow \ell \\
U & \xrightarrow{u} & R & \xrightarrow{\ell} & W & \xrightarrow{u} & C \xrightarrow{\ell} F
\end{array}$$

in which the arrows marked u are in $\mathcal{U}$ and those marked ℓ are in $\mathcal{L}$, we apply condition V3 to the middle bottom two squares and condition V4 to the middle top two squares to obtain

$$\begin{array}{ccccccc}
Q & \xrightarrow{u} & Y & \xrightarrow{u} & I & \xrightarrow{\ell} & A \xrightarrow{\ell} D \\
\downarrow u & \lrcorner & \downarrow u & \lrcorner & \downarrow u & \Rightarrow & \downarrow u \\
X & \xrightarrow{u} & Z & \xrightarrow{u} & P & \xrightarrow{\ell} & B \xrightarrow{\ell} E \\
\downarrow \ell & \Leftarrow & \downarrow \ell & \Leftarrow & \downarrow \ell & \lrcorner & \downarrow \ell \\
U & \xrightarrow{u} & R & \xrightarrow{u} & P' & \xrightarrow{\ell} & C \xrightarrow{\ell} F
\end{array}$$

which is again a fake pullback. □

As a corollary of all this we have:

Theorem 13. *Let $\mathcal{S}$ be a fake pullback ready bicategory. There is a category $\text{Spn}[\mathcal{S}]$ whose objects are those of $\mathcal{S}$, whose morphisms are isomorphism classes of spans in $\mathcal{S}$, and whose composition is defined by fake pullback.*

Remark 14. Given Remark 8, we might call $\mathcal{S}$ *proper* when the identity span provides a fake pullback of each morphism with itself. In this case, each morphism $X \xrightarrow{\mathbf{r}} Y$ in $\text{Spn}[\mathcal{S}]$ satisfies $\mathbf{r}\mathbf{r}^\circ = \mathbf{r}$ where $\mathbf{r}^\circ : Y \rightarrow X$ is the reverse span of $\mathbf{r}$.

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